

## Lecture 7

### Congruences mod Primes, Order, Primitive Roots

*Continuation of Proof of Hensel's Lemma.* By lemma,

$$f(a + tp^j) \equiv f(a) + tp^j f'(a) \pmod{p^{j+1}}$$

Now we want to have the right hand side  $\equiv 0 \pmod{p^{j+1}}$ .

$$f(a) + tp^j f'(a) \equiv 0 \pmod{p^{j+1}} \leftrightarrow tf'(a) + \frac{f(a)}{p^j} \equiv 0 \pmod{p}$$

this has a unique solution

$$t \equiv - \left( \frac{f(a)}{p^j} \frac{1}{f'(a)} \right) \pmod{p}$$

■

**Direct formula** - start with solution  $a$  of  $f(x) \equiv 0 \pmod{p}$ , and we want a solution  $\pmod{p^*}$ . Set  $a_1 = a$ .

$$a_{j+1} = a_j - f(a_j) \overline{f'(a)} \pmod{p^{j+1}}$$

where  $\overline{f'(a)}$  is an integer chosen once at the beginning of the algorithm, which only matters  $\pmod{p}$ . It's chosen such that  $\overline{f'(a)} f'(a) \equiv 1 \pmod{p}$ . Then  $f(a_j) \equiv 0 \pmod{p^j}$  for  $j \geq 1$  as long as  $f'(a) \not\equiv 0 \pmod{p}$ .

**Eg.** Solve the congruence  $x^2 \equiv -1 \pmod{125}$ . ( $f(x) = x^2 + 1$ ,  $f'(x) = 2x$ ). Mod 5:  $2^2 \equiv -1 \pmod{5}$ , so set  $a = 2$ .  $f'(a) \equiv 4 \pmod{5}$ , so can choose  $\overline{f'(a)} = -1$ .

$$\begin{aligned} a_1 &= 2 \pmod{5} \\ a_2 &= a_1 - f(a_1) \overline{f'(a)} \pmod{25} \\ &= 2 - (5)(-1) \pmod{25} \\ &= 7 \pmod{25} \\ a_3 &= a_2 - f(a_2) \overline{f'(a)} \pmod{125} \\ &= 7 - (50)(-1) \pmod{125} \\ &= 57 \pmod{125} \end{aligned}$$

**Congruences to prime modulus:** Assume that all the coefficients of  $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$  are reduced  $\pmod{p}$  and also that  $a_n \not\equiv 0 \pmod{p}$ . By dividing out by  $a_n$ , can assume that  $f(x)$  is monic (ie., highest coefficient is 1). We can also assume degree  $n$  of  $f$  is less than  $p$ . If not, can divide  $f$  by  $x^p - x$  to get

$$\begin{aligned} f(x) &= g(x)(x^p - x) + r(x) \text{ with } \deg(r(x)) < p \\ f(a) &= g(a)(a^p - a) + r(a) \equiv r(a) \pmod{p} \text{ by Fermat} \end{aligned}$$

so roots of  $f(x) \pmod{p}$  are the same as the roots of  $r(x) \pmod{p}$ .

**Theorem 28.** A congruence  $f(x) \equiv 0 \pmod p$  of degree  $n$  has at most  $n$  solutions.

*Proof.* (imitates proof that polynomial of degree  $n$  has at most  $n$  complex roots)

Induction on  $n$ : congruences of degree 0 and 1 have 0 and 1 solutions, trivially. Assume that it holds for degrees  $< n$  ( $n \geq 2$ )

If it has no roots, then we're done. Otherwise, suppose it does have a root  $\alpha$ . Dividing  $f(x)$  by  $x - \alpha$ , we get  $g(x) \in \mathbb{Z}[x]$  and a constant  $r$  such that  $f(x) = g(x)(x - \alpha) + r$ . Now if we plug in  $\alpha$  we get  $f(\alpha) = (\alpha - \alpha)g(\alpha) + r = r$ , which means that  $f(\alpha) = r$  and  $f(x) = (x - \alpha)g(x) + f(\alpha)$ .

We know that  $f(\alpha) \equiv 0 \pmod p$ . If  $\beta$  is any other root of  $f(x)$  then we plug  $\beta$  into the equation to get  $f(\beta) = (\beta - \alpha)g(\beta) + f(\alpha)$ . Mod  $p$ ,  $f(\beta) \equiv (\beta - \alpha)g(\beta) \pmod p$ , so  $0 \equiv (\beta - \alpha)g(\beta)$ . We also assume that  $\beta \neq \alpha$ , so  $g(\beta) \equiv 0 \pmod p$ .

So  $\beta$  is a root of  $g(x)$  as a solution of  $g(x) \equiv 0 \pmod p$ . We know that  $g(x)$  has degree  $n - 1$ , so by induction hypothesis  $g(x) \equiv 0 \pmod p$  has at most  $n - 1$  solutions, which by including  $\alpha$  gives  $f(x)$  at most  $n$  solutions. ■

**Corollary 29.** If  $a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 \equiv 0 \pmod p$  has more than  $n$  solutions, then all  $a_i \equiv 0 \pmod p$ .

**Theorem 30.** Let  $f(x) = x^n + a_{n-1} x^{n-1} + \dots + a_0$ . Then  $f(x) \equiv 0 \pmod p$  has exactly  $n$  distinct solutions if and only if  $f(x)$  divides  $x^p - x \pmod p$ . I.e., there exists  $g(x) \in \mathbb{Z}[x]$  such that  $f(x)g(x) = x^p - x \pmod p$  as polynomials (all coefficients mod  $p$ )

*Proof.* Suppose  $f(x)$  has  $n$  solutions. Then  $n \leq p$  because only  $p$  possible roots mod  $p$  (ie.,  $\deg(f) \leq \deg(x^p - x)$ ). Divide  $x^p - x$  by  $f(x)$  to get

$$x^p - x = f(x)g(x) + r(x), \quad \deg(r) < \deg(f) = n$$

Now note, if  $\alpha$  is a root of  $f(x) \pmod p$  then plug in to get

$$\begin{aligned} \alpha^p - \alpha &= f(\alpha)g(\alpha) + r(\alpha) \\ &\equiv 0g(\alpha) + r(\alpha) \\ &\equiv r(\alpha) \pmod p \end{aligned}$$

so  $\alpha$  must be a solution to  $r(x) \equiv 0 \pmod p$ . Since  $f(x)$  has distinct roots, we see that  $r(x) \equiv 0 \pmod p$  has  $n$  distinct solutions. But  $\deg(r) < n$ . So by corollary we must have  $r(x) \equiv 0 \pmod p$  as a polynomial (each coefficient is 0 mod  $p$ ). I.e.,  $x^p - x = f(x)g(x) \pmod p$ , and so  $f(x)$  divides  $x^p - x$ .

Now suppose  $f(x) | x^p - x \pmod p$ . Write  $x^p - x \equiv f(x)g(x) \pmod p$ , where  $f(x)$  is a monic of degree  $n$  and  $g(x)$  is a monic of degree  $p - n$ . We want to show that  $f(x)$  has  $n$  distinct solutions.

By previous theorem,  $g(x)$  has at most  $p - n$  roots mod  $p$ . If  $\alpha \in 0, 1, \dots, p - 1$  is not a root of  $g(x) \bmod p$  then  $\alpha^p - \alpha \equiv f(\alpha)g(\alpha) \bmod p$ , which by Fermat  $\equiv 0$ . Since  $g(\alpha) \not\equiv 0 \bmod p$ ,  $f(\alpha) \equiv 0 \bmod p$ . So since there are at least  $p - (p - n)$  such  $\alpha$ , we see that  $f(x)$  has at least  $n$  distinct roots mod  $p$ . By the theorem,  $f(x)$  has at most  $n$  roots mod  $p \Rightarrow f(x)$  has exactly  $n$  distinct roots mod  $p$ . ■

**Corollary 31.** If  $d|p - 1$  then  $x^d \equiv 1 \bmod p$  has exactly  $d$  distinct solutions mod  $p$ .

*Proof.*  $d|p - 1$ , so  $x^{d-1} - 1 | x^{p-1} - 1$  as polynomials.  $p - 1 = kd$ , so  $x^{kd} - 1 = (x^d - 1)(x^{(k-1)d} + \dots + 1)$ . So  $x^d - 1 | x(x^{p-1} - 1) = x^p - x$ . So has  $d$  solutions. ■

**Corollary 32.** Another proof of Wilson's Theorem

*Proof.* Let  $p$  be an odd prime. Let  $f(x) = x(x - 1)(x - 2) \dots (x - p + 1)$ . This has  $\deg p$  and  $p$  solutions mod  $p$ , so it must divide  $x^p - x \bmod p$ . Both polynomials are monic of the same degree ( $p$ ), so must be equal mod  $p$ .

$$x(x - 1) \dots (x - (p - 1)) \equiv x^p - x \bmod p$$

Coefficient of  $x$  on the LHS is just  $(-1)(-2) \dots (-(p - 1)) = (-1)^{p-1}(p - 1)! = (p - 1)!$  since  $p$  is odd, and so  $(p - 1)! \equiv -1 \bmod p$  (coefficient on RHS). ■

This tells us much more as well - eg.,  $1 + 2 + \dots + p - 1 \equiv 0 \bmod p$  for  $p \geq 3$ , and  $(1)(2) + (1)(3) + \dots + (2)(3) \dots + (p - 1)(p - 2) \equiv 0 \bmod p$  for  $p \geq 5$ .

If we have a product  $f(x) = (x - \alpha_1) \dots (x - \alpha_n)$  then  $f(x) = x^n - \sigma_1 x^{n-1} + \sigma_2 x^{n-2} + \dots + (-1)^n \sigma_n$ .  $\sigma_i$  are elementary symmetric polynomials.

$$\begin{aligned}\sigma_1 &= \sum \alpha_i \\ \sigma_2 &= \sum_{i < j} \alpha_i \alpha_j \\ \sigma_k &= \sum (\text{all products of } k \text{ roots } \alpha_i)\end{aligned}$$

**Question** - We know by Euler that if  $(n, 35) = 1$ , then  $n^{\phi(35)} = n^{24} \equiv 1 \bmod 35$ . Can 24 be replaced by something smaller? I.e., what's the smallest positive integer  $N$  such that if  $(n, 35) = 1$  then  $n^N \equiv 1 \bmod 35$ .

**(Definition) Order:** If  $(a, m) = 1$  and  $h$  is the smallest positive integer such that  $a^h \equiv 1 \bmod m$  then say  $h$  is the **order** of  $a \bmod m$ . Written as  $h = \text{ord}_m(a)$ .

**Lemma 33.** Let  $h = \text{ord}_m(a)$ . The set of integers  $k$  such that  $a^k \equiv 1 \bmod m$  is exactly the set of multiples of  $h$ .

*Proof.*  $a^{rh} \equiv (a^h)^r \equiv 1^r \equiv 1 \pmod{m}$ . Suppose we have  $k$  such that  $a^k \equiv 1 \pmod{m}$ . Want to show  $h|k$ . Write  $k = hq + r$  where  $0 \leq r < h$ .  $1 \equiv a^k = a^{hq+r} = a^{hq}a^r \equiv 1a^r \equiv a^r \pmod{m}$ , so  $a^r \equiv 1 \pmod{m}$ . But  $r < h$ . So if  $r > 0$ , contradicts minimality of  $h$ , which means that  $r = 0$ , and  $k$  is multiple of  $h$ . ■

**Lemma 34.** If  $h = \text{ord}_m(a)$  then  $a^k$  has order  $\frac{k}{(k,h)} \pmod{m}$ .

*Proof.*

$$\begin{aligned} a^{kj} &\equiv 1 \pmod{m} \\ &\Leftrightarrow h|kj \\ &\Leftrightarrow \frac{h}{(h,k)} \mid \frac{k}{(h,k)}j \\ &\Leftrightarrow \frac{h}{(h,k)} \mid j \end{aligned}$$

So smallest such positive  $j = \frac{h}{(h,k)}$ . ■

**Lemma 35.** If  $a$  has order  $h \pmod{m}$  and  $b$  has order  $k \pmod{m}$ , and  $(h, k) = 1$ , then  $ab$  has order  $hk \pmod{m}$ .

*Proof.* We know

$$\begin{aligned} (ab)^{hk} &\equiv (a^h)^k (b^k)^h \\ &\equiv 1^k 1^h \\ &\equiv 1 \pmod{m} \end{aligned}$$

Conversely suppose that  $r = \text{ord}_m(ab)$ .

$$\begin{aligned} (ab)^r &\equiv 1 \pmod{m} \\ (ab)^{rh} &\equiv 1 \pmod{m} \\ (a^h)^r b^{rh} &\equiv 1 \pmod{m} \\ b^{rh} &\equiv 1 \pmod{m} \end{aligned}$$

so  $k|rh \Rightarrow k|r$  (since  $(k, h) = 1$ ), and similarly  $h|r$ . So  $hk|r$ , and so  $hk = \text{ord}_m(ab)$ . ■

**(Definition) Primitive Root:** If  $a$  has order  $\phi(m) \pmod{m}$ , we say that  $a$  is a **primitive root** mod  $m$ .

**Eg.** mod 7:

|   |           |   |                           |
|---|-----------|---|---------------------------|
| 1 | has order | 1 |                           |
| 2 | has order | 3 | $(2^3 \equiv 1 \pmod{7})$ |
| 3 | has order | 6 | ✓ $(\phi(7) = 6)$         |
| 4 | has order | 3 |                           |
| 5 | has order | 6 | ✓ $(\phi(7) = 6)$         |
| 6 | has order | 2 |                           |

**Lemma 36.** *Let  $p$  be prime and suppose  $q^e \parallel p - 1$  for some other prime  $q$ . Then there's an element mod  $p$  of order  $q^e$ .*

**Assuming Lemma...**

$$p - 1 = q_1^{e_1} q_2^{e_2} \dots q_r^{e_r}$$

Lemma says that  $\exists g_1$  with  $\text{ord}_p(g_1) = q_1^{e_1}$ ,  $g_2$  with  $\text{ord}_p(g_2) = q_2^{e_2}$ , etc. Set  $g = g_1 g_2 \dots g_r$ . So by previous lemma above,  $g$  has order  $q_1^{e_1} q_2^{e_2} \dots q_r^{e_r} = p - 1$  because all  $q_i$  are coprime in pairs.  $p - 1 = \phi(p)$ , so  $g$  is a primitive root mod  $p$ .

*Proof.* Consider solutions of  $x^{q^e} \equiv 1 \pmod{p}$ . Because  $q^e \mid p - 1$ ,  $x^{q^e} - 1$  has exactly  $q^e$  roots mod  $p$ . If  $\alpha$  is any such root, then  $\text{ord}_p(\alpha)$  must divide  $q^e$ .

So if it's not equal to  $q^e$ , it must divide  $q^{e-1}$ . Then  $\alpha$  would have to be root of  $x^{q^{e-1}} - 1 \equiv 0 \pmod{p}$ , which has exactly  $q^{e-1}$  solutions. Since  $q^e - q^{e-1} > 0$ , there exists  $\alpha$  such that  $\text{ord}_p(\alpha) = q^e$ . ■

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18.781 Theory of Numbers  
Spring 2012

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